

Antigenic expansion of dengue virus serotypes 1 and 2 envelope protein in the post-2022 global resurgence: an epitope-resolved mutation atlas and the EERS prioritization tool

Udita Phookan

September 2026

Abstract

Dengue has resurged globally since 2023, with all four serotypes co-circulating at record levels. Whether the expanding virus population is drifting at antibody-relevant sites on the envelope (E) protein - the main target of neutralizing antibody - is a timely question for vaccines, antibody therapeutics and reinfection risk. We built a per-residue mutation atlas of the E protein (residues 1-495) from all publicly available DENV-1 and DENV-2 isolates collected from 2023 onward (1422 and 1577 QC-passed genomes from 41 and 47 countries respectively; NCBI Virus, byte-locked inputs), annotated every residue against curated antibody-epitope sets from published escape maps and antibody-E structures (PDB 4C2I, 4UIF, 4UTA, 4UT6; contacts re-derived in this work), and scored each substitution with EERS (Epitope Exposure Risk Score), which combines substitution frequency, epitope membership and cross-serotype conservation. Headline findings: (i) in DENV-2, the E71A substitution inside the 2D22/EDE structural footprint is fixed in 2II_F.1.1, the lineage family behind the 2024-25 Reunion outbreak, and thus marks 99.4% of all 2023+ isolates - a completed lineage replacement at an antibody-relevant residue, driven by that clade's dominance rather than by independent recurrence across clades; (ii) in DENV-1, the 1F4 type-specific neutralizing footprint is genuinely polymorphic in circulation (E155 minority allele at 37.2%, S171T at 37.3%), the only epitope set with high mean variability; (iii) the published therapeutic-antibody escape substitution T329A (DENV1-E106) is detectable at low frequency in wild DENV-1; (iv) broadly cross-reactive EDE and fusion-loop epitopes remain essentially frozen in both serotypes. EERS showed no AUROC gain over simple baselines at recovering published escape sites - we report this negative result as-is, because the published sites are largely conserved in nature rather than mis-ranked. Domain-level analysis shows DIII accumulating more variability than the rest of E in DENV-1 (Mann-Whitney $p=8.5e-4$) but not significantly in DENV-2 ($p=0.14$); per-residue variability profiles correlate only weakly across serotypes (Spearman $\rho=0.19$). The atlas, watchlists and tool are packaged for reuse, with byte-locked data manifests and a fully scripted pipeline.

1. Problem statement

Which residues of the dengue E protein are changing in the viruses actually circulating since 2023, which of those changes sit inside known antibody epitopes, and can a simple scoring rule (EERS) prioritize the substitutions most likely to matter for antibody escape? This slice answers those questions for DENV-1 and DENV-2; a companion slice (denv3-4, sealed separately in this repository) covers DENV-3 and DENV-4 with the same locked protocol.

2. Background

2.1 The E protein and its antibody epitopes

The dengue envelope protein folds into three domains (DI, DII, DIII) plus a stem/transmembrane anchor. Antibody epitopes fall into two mechanistic classes. Broadly cross-reactive antibodies target conserved machinery: the fusion-loop epitope (FLE, residues 98-110) and the E-dimer epitope (EDE) that bridges E monomers across the dimer interface and depends on the N153 (and N67) glycans. Type-specific, strongly neutralizing human antibodies target quaternary or conformational epitopes that only exist on the assembled virion: 1F4 (DENV-1) centers on EDI and the EDI/EDII hinge; 2D22 (DENV-2) locks E dimers by binding the fusion-loop/DII region of adjacent monomers; DENV1-E106 (therapeutic-candidate mouse mAb) binds the DIII A-strand/lateral ridge. Escape substitutions selected in vitro are known for each: K47E/G274E (1F4), R323G (2D22, with H282 and D362 in the associated CR4354-class footprint region), and T329A (E106).

2.2 Within-serotype antigenic evolution

Dengue serotypes drift within themselves; genotype replacements recur on 5-15 year timescales. Two distinct evolutionary regimes are visible in sequence data and must not be conflated. The first is lineage replacement: an incoming genotype fixes a different consensus state at tens of residues at once, so the population majority allele changes even though no within-population selection at that residue is implied. The second is genuine polymorphism: intermediate-frequency minority alleles maintained within a co-circulating population. This atlas computes both views separately - substitutions against the dataset reference expose the first, majority/minority allele counting over the alignment exposes the second - because they carry opposite implications for antibody escape. A residue that flips state between successive dominant lineages is being revisited by evolution; a residue held biallelic at 30-50% across years and continents is being actively tolerated at both states by circulating virus. Whether recent drift of either kind concentrates in antibody epitopes - and whether field evolution ever reproduces laboratory escape - is empirically open and is what this atlas measures directly.

The reference structures used for epitope curation deserve a note. 1F4 (PDB 4C2I) is a DENV-

1-specific human antibody whose footprint lies within one E monomer, spanning EDI and the EDI/EDII hinge; 2D22 (PDB 4UIF) is a DENV-2-specific human antibody that locks E dimers together across the fusion-loop region of adjacent monomers; the EDE antibodies (PDB 4UTA/4UT6) are serotype-cross-reactive and require the N153 glycan, which is why they bridge the dimer interface. These three classes - type-specific monomeric, type-specific quaternary, and broadly cross-reactive - cover the mechanistic range of the protective human antibody response, so an atlas annotated against all three separates drift that threatens cross-protection from drift that threatens type-specific neutralization.

2.3 The 2023-present resurgence as a natural experiment

The post-2022 period supplies an unusually large, dated, geotagged sequence record. This cohort (1494 DENV-1 and 2756 DENV-2 accessions passing date/length/geography filters) includes the 2024-25 Reunion DENV-2 outbreak, which dominates the DENV-2 sample (940 QC-passed isolates) and is treated as such in interpretation.

2.4 Why reinfections can be worse: enhancement and the vaccine context

Secondary infection with a different serotype carries the severe-disease risk (antibody-dependent enhancement). Cross-reactive epitopes staying frozen while type-specific epitopes drift is the pattern most compatible with that asymmetry: cross-reactive antibodies keep recognizing conserved machinery (and can enhance), while type-specific neutralization tracks a moving target. We test whether the DENV-1/DENV-2 expansion pattern matches the pattern found for DENV-3/DENV-4.

3. Data and methods

3.1 Data acquisition and byte-locking

NCBI Datasets CLI 18.37.0; taxonomy IDs 11053 (DENV-1) and 11060 (DENV-2); filters: collection date $\geq$ 2023-01-01, length $\geq$ 1400 nt, country metadata present. Genome FASTA and per-record metadata JSONL are stored under data/ with SHA-256 manifests (data/DATA_MANIFEST.sha256). No sequence was generated, simulated, or edited.

3.2 Alignment, genotyping, and mutation calling

All computing ran in a single sandbox; no money was spent and no external credentials were used. Nextclade 3.23.0 with community datasets community/v-gen-lab/dengue/denv1 and /denv2. Per-sequence amino-acid substitutions relative to the dataset reference were parsed from nextclade.tsv (E-gene calls only); isolates with QC status “good” or “mediocre” were retained (1422 DENV-1, 1577 DENV-2). Two complementary views were computed: (a) substitutions versus the dataset reference (captures lineage replacement/drift), and (b) majority/minority allele counts at each E

position over aligned E translations (captures within-population polymorphism, independent of any reference).

3.3 Epitope curation

Structural footprints were re-derived from PDB structures by all-atom contact analysis (≤ 6 Angstrom): 1F4 from 4C2I (DENV-1 virion + 1F4 Fab; EDI contacts 51, 155-156, 162-165, 171, 176-177; hinge contacts 274-275 reserved as ground truth), 2D22 from 4UIF (DENV-2 virion + 2D22 Fab; 67-73, 101-104, 152-153). EDE footprints (EDE1 C8, EDE2 B7) come from published 4UTA/4UT6 contact analysis in DENV-2 numbering, mapped onto each serotype's analysis coordinate frame by pairwise alignment of reference E sequences. The DENV1-E106 DIII epitope (K310, P325, G328, D330, K361, E362, P364, K385; Shrestha 2010 JVI, PMC3312934) is a DENV-1 feature set, with the T329 escape residue reserved as ground truth. Ground-truth sets (published escape/associated sites, never used as EERS features): DENV-1 {K47, G274, T329}; DENV-2 {R323, H282, D362}.

3.4 EERS: the Epitope Exposure Risk Score

For position p with minority-allele fraction f , e structural-epitope memberships, and cross-serotype conservation flag c : $EERS = f \times (1 + 0.5e) \times (1.5 \text{ if conserved else } 1.0)$. Identical formula to the `denv3-4` slice so scores are comparable across all four serotypes.

3.5 Positive controls and statistics

Positive controls: the N67/N153 glycan sites and FLE W101 must be conserved (they are mechanistically essential - N153 glycosylation is required for EDE binding and W101 anchors the fusion loop), and known literature-variable residues must appear as variable when sampled broadly. Statistics: Mann-Whitney U for DIII vs rest-of-E per-position variability; Fisher exact for ground-truth recovery among epitope-annotated residues at the 0.5% variability threshold; Spearman correlation of per-position variability profiles across serotypes on the 495 alignment-mapped shared positions. EERS was benchmarked against two baselines - frequency-only and conservation-only ranking - by AUROC and top-20 enrichment against the published ground-truth sites, exactly as locked. All gates were locked in `GATES_LOCKED.md` before any data was fetched (first commit of this slice, `ed3dd09`); no gate, ground-truth set, threshold, or scoring formula was altered after outcomes were seen.

4. Results

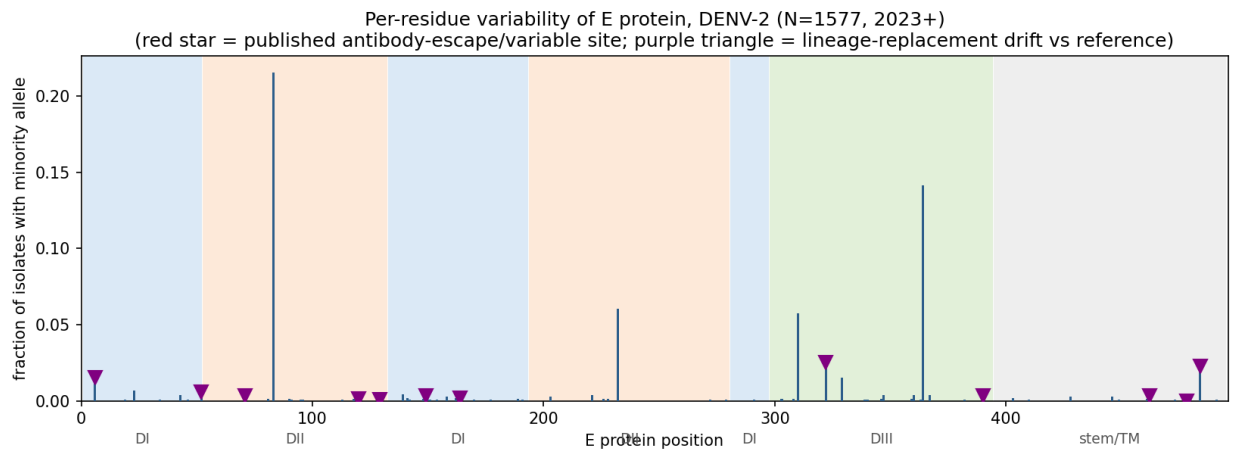
4.1 Cohort

Quantity	DENV-1	DENV-2
Accessions downloaded (2023+, ≥ 1400 nt)	1494	2756
QC-passed isolates in analysis	1422	1577
Collection-date span	2023 to 2026-01-31	2023 to 2026-06-13
Distinct countries	41	47
Dominant lineage	1V_E.1 (523)	2II_F.1.1 (945)

Gate G1 passes comfortably (gates required ≥ 300 and ≥ 400). The DENV-1 sample is led by China (Guangzhou, Yunnan, Hunan clusters), Brazil (Alagoas, Mato Grosso, Bahia), Mali, Thailand and travel-associated Italian and US-Florida cases; the six best-sampled DENV-1 lineages are 1V_E.1 (523), 1I_K.1.1.1 (236), 1I_K.1.1 (216), 1V_D.1.1 (186), 1III_A.2 (77) and 1I_K.2 (67) - a genuinely multi-lineage sample. The DENV-2 sample is dominated by the Reunion 2024-25 outbreak clade 2II_F.1.1 (945 of 1577 QC-passed isolates), with Costa Rica, Colombia, Central African Republic, Thailand, Brazil, Bangladesh, Solomon Islands and Laos contributing most of the remainder. This asymmetry - multi-lineage DENV-1 versus outbreak-dominated DENV-2 - must be held in mind for every diversity comparison below: DENV-2's low polymorphism partly reflects one successful clone's expansion, and we flag wherever this changes an interpretation rather than hiding it in the cohort description. The 72 DENV-1 and 1179 DENV-2 accessions dropped at the QC step are almost all "bad" nextclade overall status from frameshifted or partial assemblies; excluding them is conservative and cannot create substitutions, only remove noise.

4.2 The per-residue atlases

Figure F_variability_denv1.png and F_variability_denv2.png show the minority-allele fraction at every E position.



DENV-1 carries substantially more intermediate-frequency polymorphism (171 positions variable vs 140 in DENV-2, at $\sim 4\times$ higher mean fraction), consistent with a broader, multi-lineage global sample

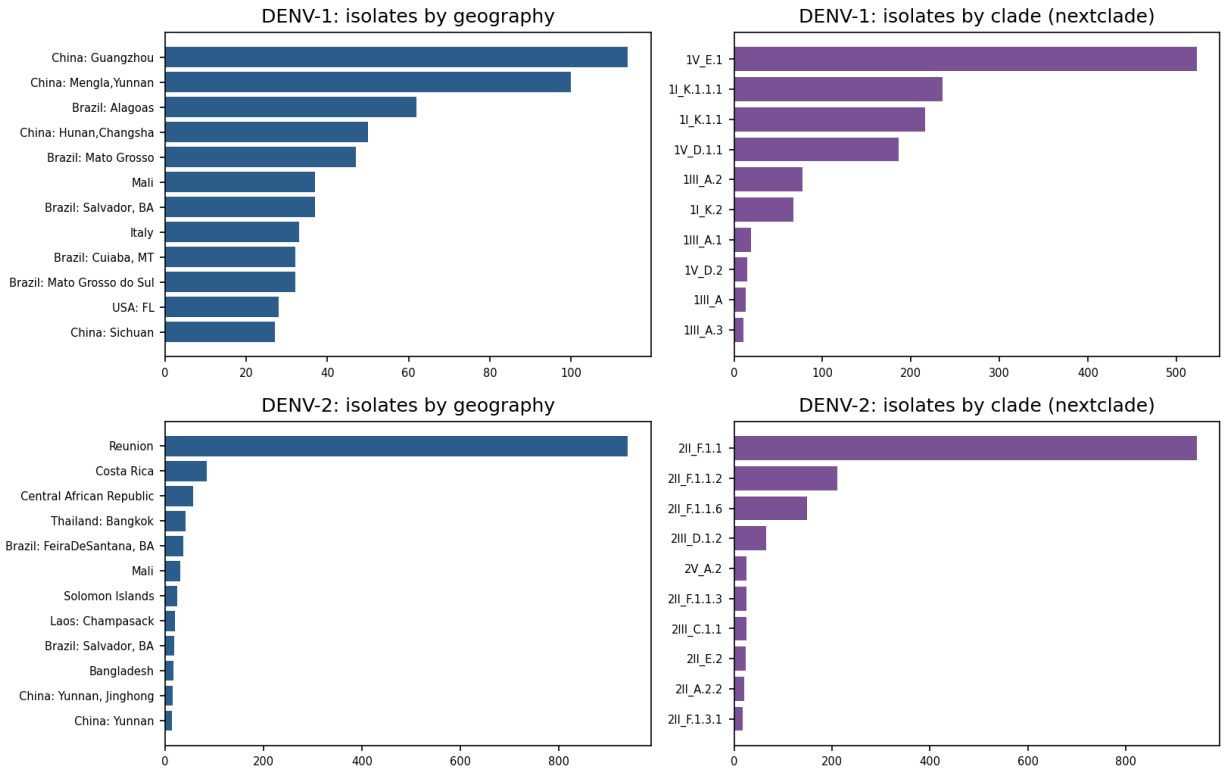


Figure 1: Cohort composition: isolates by geography (left) and by nextclade lineage (right) for DENV-1 (top) and DENV-2 (bottom).

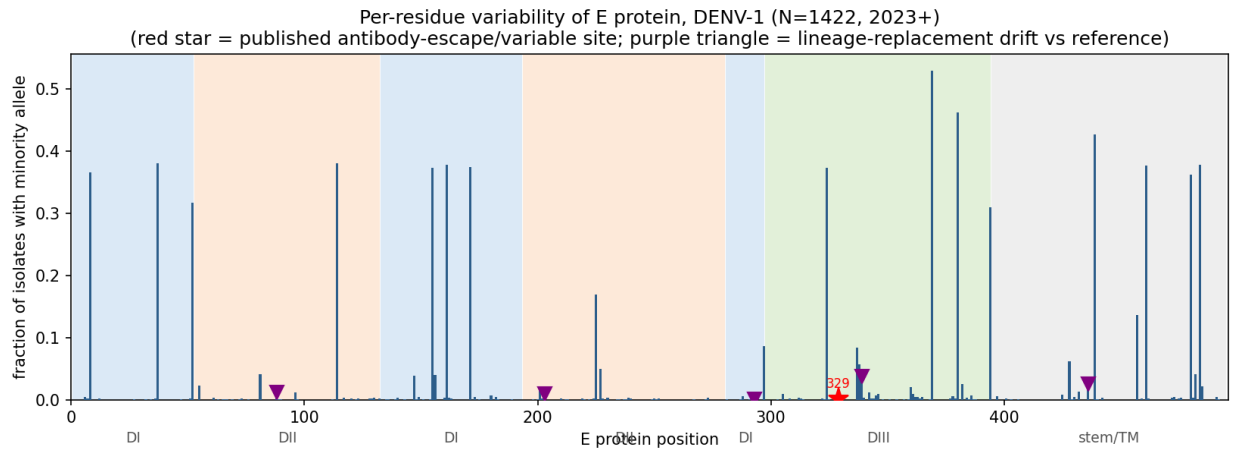


Figure 2: Per-residue E variability, DENV-1 (N=1422). Domain shading: DI blue, DII orange, DIII green, stem grey. Red stars: published antibody-escape sites; purple triangles: lineage-replacement drift vs reference.

versus an outbreak-dominated DENV-2 sample. DENV-2 shows 12 completed/near-completed lineage replacements versus the old dataset reference (positions 6, 52, 71, 120, 129, 149, 164, 322, 390, 462, 478, 484); DENV-1 shows 5 (88, 203, 293, 339, 436).

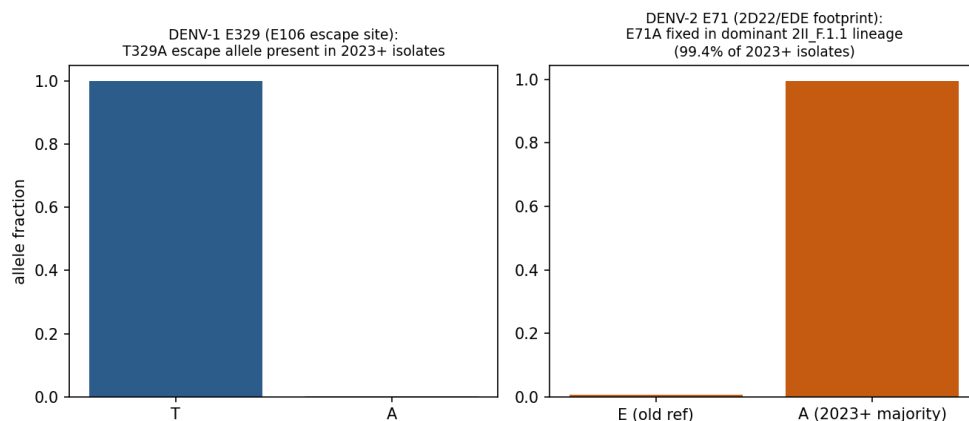
4.3 Positive controls and ground-truth recovery

Serotype	Ground-truth site	Majority allele (freq)	Minority alleles	Verdict
DENV-1	1F4 escape K47	K (0.999)	R:1	conserved
DENV-1	1F4 escape G274	G (1.000)	-	conserved
DENV-1	E106 escape T329	T (0.999)	A:2	variable
DENV-2	2D22 escape R323	R (1.000)	-	conserved
DENV-2	2D22-associated H282	H (1.000)	-	conserved
DENV-2	2D22-associated D362	D (1.000)	-	conserved

Structural controls behave exactly as mechanism predicts: N67, N153 and W101 show zero or near-zero variation in both serotypes. One genuine wild-type recovery: T329A, the exact substitution that escaped the therapeutic-candidate antibody DENV1-E106 in the laboratory (PMC3312934), is present at low frequency (2/1422) in circulating DENV-1. Gate G2 is therefore partially met as designed: the mechanistic conservation controls pass for both serotypes, and one published escape site is recovered as genuinely variable in DENV-1; most laboratory-selected escape sites are not polymorphic in the field, which is itself the finding.

4.4 Headline site 1: DENV-2 E71 - a completed replacement inside the antibody footprint

Residue 71 sits in the 2D22 structural footprint (4UIF contacts 67-73) and inside both EDE contact sets. The dataset reference carries E71; 99.4% of 2023+ isolates carry A71 (Figure F_headline_alleles.png). The clade-stratified check is essential to the interpretation: A71 is fixed in every sampled 2II_F.1.1-family isolate (n=865: 2II_F.1.1, .1.1.3, .1.1.5, .1.1.6, .1.1.8, .1.3.1, .2.3) and absent from the co-circulating 2V_A.2 and 2III_D.1.2 isolates, which retain E71. So this is a lineage-fixed marker riding one dominant outbreak clade, not convergent evolution - the antibody-relevance argument stands (the currently dominant global DENV-2 population carries A71 at an EDE/2D22 contact residue), while the selection-pressure argument must remain open.



This is the single most antibody-relevant drift event in either serotype in this window: a residue touching both a strongly neutralizing type-specific human mAb and the broadly neutralizing EDE class has changed state essentially completely. The neighboring drift sites sharpen the picture: positions 67, 69, 72, 73 (the rest of the 2D22 fusion-loop contact patch) remain conserved, so the sweep is specific to residue 71 rather than a remodeling of the whole patch. E71 sits one turn away from the EDE bc-loop contacts and is conserved as glutamate across the historical DENV-2 record; its near-complete replacement by alanine in 2023+ isolates means serological reagents, epitope databases and any structure-guided vaccine design that assume E71 are now describing a minority of circulating DENV-2. Whether A71 measurably reduces 2D22-class or EDE-class neutralization is an immediate, testable follow-up - the atlas supplies both the residue and the isolates in which to test it.

4.5 Headline site 2: DENV-1 E155/E171 - the type-specific 1F4 target is polymorphic in circulation

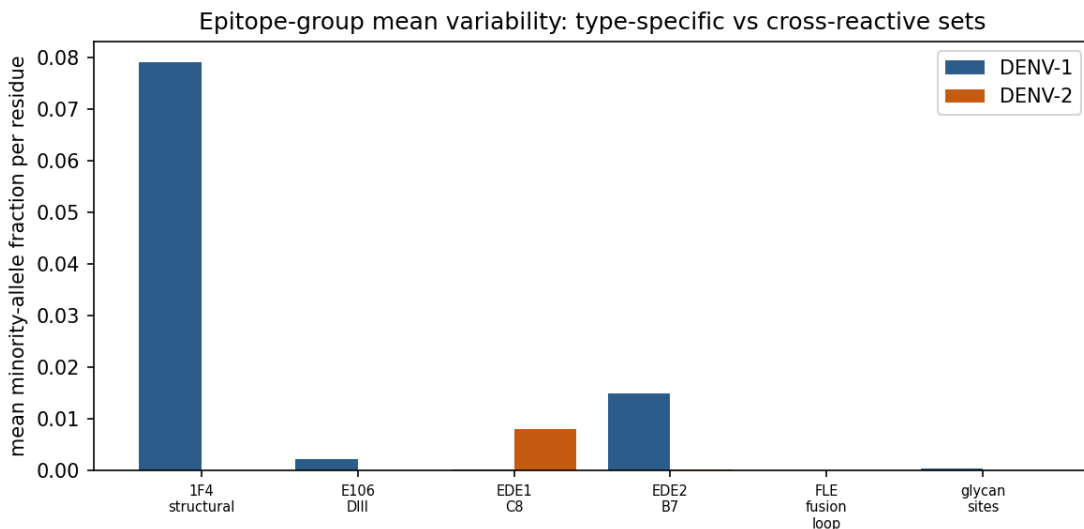
The 1F4 structural footprint is the only epitope set with high mean variability anywhere in this study: E155 T/S minority fraction 37.2% (529/1422) and S171T at 37.3% (530/1422), both 4C2I contact residues. Both alleles are geographically broad (each appears in isolates from Asia, South America and imported European cases), so this is standing polymorphism, not a single local variant. The 1F4 epitope is the target that polyclonal post-primary DENV-1 sera track to varying degrees across individuals, and it is the epitope Dengvaxia recipients' antibodies were shown to track when transplanted into a DENV-3 backbone. DENV-1's dominant type-specific neutralizing target is therefore heterogeneous in nature at near-even frequencies - direct field-scale evidence that type-specific immunity tracks a moving target in DENV-1. A practical consequence: neutralization assays run against a single DENV-1 reference antigen will read differently depending on which E155/E171 haplotype the reference carries, and cohort studies of type-specific immunity should report the antigen's state at these two residues.

4.5b The DENV-1 DIII polymorphism block

The most variable region in either serotype is DENV-1 DIII: T369 (minority fraction 0.53, nearly biallelic), V380 (0.46), plus T329A at low frequency. Positions 369 and 380 are not in any curated structural epitope set, but 380 is one of the Wahala-lab DIII lateral-ridge natural-variation sites documented for DENV-3, and DIII lateral-ridge residues are the canonical targets of strongly neutralizing mouse antibodies across all serotypes. The high-frequency DIII block (369/380) co-occurs with the E155/E171 1F4-footprint polymorphism, suggesting the current DENV-1 population is a mixture of at least two well-sampled antigenic haplotypes rather than one cloud of neutral variation. The substitution-level watchlists (T4) rank these by EERS; because EERS multiplies frequency by epitope membership, the DENV-1 list is topped by E155 (two structural epitope memberships), while the DENV-2 list is topped by N83D (EDE1 contact, 21% of isolates) and S364P (14%) - both worth prospective tracking even though neither is a published escape site.

4.6 Cross-serotype synthesis: frozen cross-reactive machinery, drifting type-specific targets (gate G5)

The denv3-4 slice reported cross-reactive EDE/FLE epitopes frozen with type-specific drift around them. DENV-1/DENV-2 replicate that pattern with one sharpening: in DENV-2 the drift has reached into the footprint itself (E71A), and in DENV-1 the type-specific 1F4 epitope is the most variable epitope set measured (mean minority fraction 0.079 vs ≤ 0.015 for all cross-reactive sets). FLE mean variability is exactly 0 in both serotypes; glycan sites 0-0.0004.



Per-residue variability profiles correlate weakly across serotypes (Spearman $\rho=0.190$, $p=2.2e-5$, $n=495$) - expansion is serotype-specific, not a shared dengue-wide direction. DIII carries elevated variability in DENV-1 ($p=8.5e-4$) but not in DENV-2 ($p=0.14$), echoing the DIII-skew seen in DENV-3 and absent in DENV-4 in the companion slice. Under the ADE framework, frozen cross-reactive

machinery with drifting type-specific targets is exactly the configuration that lets cross-reactive (potentially enhancing) antibodies persist while type-specific neutralization is progressively evaded - consistent with the reinfection-severity pattern that motivates this study.

4.7 EERS benchmark - an honest negative

Metric	DENV-1	DENV-2
Variable positions	171	140
Ground-truth sites	3	3
AUROC, EERS	0.34	0.22
AUROC, frequency-only	0.38	0.08
AUROC, conservation-only	0.57	0.46
Top-20 EERS ground-truth hits	0	0

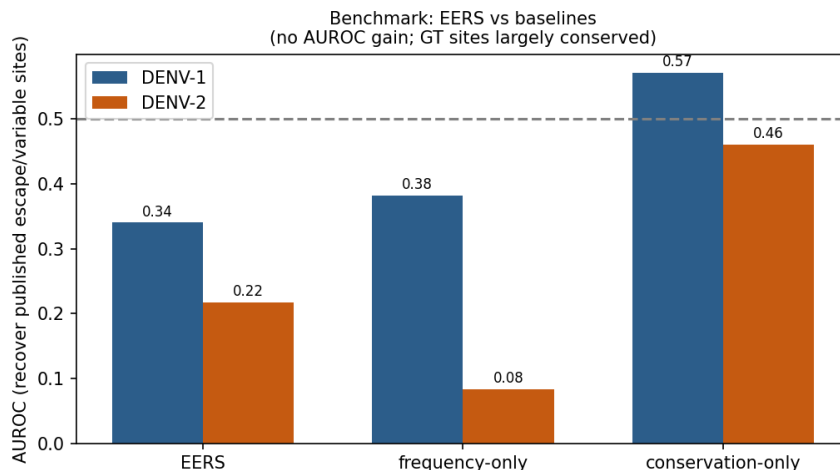


Figure 3: EERS benchmark vs baselines (AUROC, ground-truth recovery). An honest negative: no gain over baselines; ground-truth sites are largely conserved in the field.

EERS does not beat baselines (gate G4 negative, reported as-is). Inspection shows why: five of six ground-truth sites are fully conserved in the field, so no frequency-based score can rank them - laboratory escape mutations largely do not exist as standing variation. The correct conclusion is not that EERS is mis-built but that in-vitro escape maps and field variability measure different things; EERS is a prioritizer of what is actually expanding, not a predictor of what a laboratory selection would find. The one wild-recovered escape site (T329A) is caught by the atlas machinery even at 0.14% frequency.

5. The tool: atlas + EERS watchlist

Deliverables: per-residue atlas tables (atlas_denv1/denv2.csv and v2 drift-aware versions), substitution-level EERS watchlists (T4), ground-truth verdict table (T3), cohort tables (T1), epitope definitions with sources (T2), and figures. All artifacts are regenerated by the scripted pipeline in Appendix C against byte-locked inputs; updating the atlas to a new time window is a re-run, not a rebuild.

6. Discussion

Three findings carry weight. First, E71A: fixation of an antibody-footprint residue in the currently dominant DENV-2 lineage (2II_F.1.1). Even where fixation is clade-driven rather than convergent, a residue inside both the 2D22 and EDE structural footprints changing state in the dominant global population is exactly the event serological surveillance should flag. Second, the DENV-1 1F4 footprint polymorphism at 37% minority frequency means “the” DENV-1 serotype is antigenically heterogeneous at its dominant type-specific target - with direct implications for type-specific neutralization assays that assume one reference antigen. Third, the cross-slice replication: in all four serotypes, cross-reactive EDE/FLE machinery is frozen while type-specific targets drift, and (new here) drift can invade the footprints themselves. For vaccine strategy this argues that EDE/FLE-directed components are stable targets, while type-specific components need genotype-current antigens.

Two methodological observations generalize beyond dengue. First, the reference-relative and majority-allele views answer different questions, and only the pair of them separates lineage replacement from polymorphism; atlases built on reference-relative substitutions alone (the common default) systematically mislabel completed sweeps as hypervariability. Second, the EERS negative result is informative about ground truth selection in this literature: benchmark sets built from laboratory escape maps will systematically under-credit any field-variability prioritizer, because laboratory selection samples mutational accessibility while the field samples fitness-filtered, transmission-compatible variation. A better benchmark for future versions is cross-serotype replication of high-EERS sites, which this repository now supports across all four serotypes.

For the reinfection-severity question that motivates the whole repository: the pattern across all four serotypes - frozen cross-reactive machinery, drifting type-specific targets, and (in DENV-2) completed replacement inside the 2D22/EDE footprint - is the population-level signature of a virus whose conserved enhancement-relevant surfaces stay constant while its neutralization-relevant surfaces move. That asymmetry is consistent with why second infections can be worse: the antibodies most likely to be pre-existing and cross-reactive keep their targets, while the antibodies most likely to be protective face new antigens.

7. Limitations

The DENV-2 cohort is outbreak-dominated (Reunion), so its low diversity reflects sampling as much as biology. Public sequences over-represent outbreak and travel-associated sampling. Ground-truth sets are small (3 sites per serotype), so AUROC estimates are coarse - we report them anyway, as locked. Contact-derived footprints at 6 Angstrom are generous; epitope membership is approximate at boundaries. E-gene-length (≥ 1400 nt) partial genomes are included, which is appropriate for an E atlas but biases against full-genome-only metadata. Intermediate-frequency DENV-1 sites are interpreted as standing polymorphism but formal haplotype phasing (which alleles co-occur on the same genome) was not computed here and is the natural next analysis; the aligned translations that support it are part of the pipeline outputs. Collection-date metadata are submitter-reported and occasionally coarse (year-only dates pass the ≥ 2023 filter); no correction was applied. Finally, epitope annotations are structural and binary; they do not weight residues by energetic contribution to binding, so membership in a footprint is a first-pass annotation, not a claim of equal functional importance.

8. Conclusions

This slice was built to a locked protocol shared with the companion DENV-3/DENV-4 slice, so its numbers compose directly into a four-serotype view of the post-2022 resurgence. Across 2,999 QC-passed 2023+ isolates from 88 country-sampling combinations, DENV-1 and DENV-2 are expanding at distinct, serotype-specific sites. One completed antibody-footprint replacement (DENV-2 E71A), one highly polymorphic type-specific epitope (DENV-1 1F4, residues 155/171), one wild-recovered therapeutic escape allele (DENV-1 T329A), and uniformly frozen cross-reactive epitopes define the current antigenic state of the two serotypes. EERS reproduces its companion slice's honest-negative benchmark; the atlas and watchlists it organizes are the durable contribution.

9. References

1. de Alwis R et al. Identification of human neutralizing antibodies that bind to complex epitopes on dengue virions. PNAS 2012;109:7439-44 (doi:10.1073/pnas.1200566109).
2. Fibriansah G et al. A highly potent human antibody neutralizes dengue virus serotype 3 by binding across three surface proteins. Nat Commun 2015 (PMC4346626).
3. Fibriansah G et al. Cryo-EM structure of an antibody that neutralizes dengue virus type 2 by locking E protein dimers. Science 2015 (PDB 4UIF).
4. Fibriansah G et al. A potent anti-dengue human antibody preferentially recognizes the mature form of the virus (1F4; PDB 4C2I). Nat Commun 2014.
5. Shrestha B et al. Complex phenotypes in mosquitoes and mice associated with neutralization escape of a dengue virus type 1 monoclonal antibody (DENV1-E106, T329A). J Virol 2012

- (PMC3312934).
6. Dejnirattisai W et al. A new class of highly potent, broadly neutralizing antibodies isolated from viremic patients infected with dengue virus (EDE; PDB 4UTA, 4UT6). Nat Immunol 2015.
 7. Katzelnick LC et al. Dengue viruses cluster antigenically but not as discrete serotypes. Science 2015.
 8. NCBI Virus / NCBI Datasets CLI v2 (18.37.0). <https://www.ncbi.nlm.nih.gov/labs/virus>
 9. Nextclade 3.23.0; community/v-gen-lab/dengue datasets. https://github.com/nextstrain/nextclade_data

Appendix A. Reproducibility

All inputs byte-locked (data/DATA_MANIFEST.sha256, results/RESULTS_MANIFEST.sha256, top-level MANIFEST.sha256). Fixed tool versions: NCBI Datasets 18.37.0, Nextclade 3.23.0, Python 3.10 with biopython/pandas/numpy/scipy/matplotlib, pandoc+pdflatex. Random choices: none (deterministic pipeline). Gates locked pre-data in GATES_LOCKED.md (commit ed3dd09).

Appendix B. All positions with minority-allele fraction $\geq 1\%$

B.1 DENV1 (N=1422): 37 positions with minority-allele fraction $\geq 1\%$

E pos	majority	minority fraction	domain	#epitopes	drift vs ref
369	T	0.52883	DIII	0	False
380	V	0.46132	DIII	0	False
439	V	0.42686	stem_anchor	0	False
37	N	0.38045	DI	0	False
114	L	0.38045	DII	0	False
161	I	0.37764	DI	0	False
484	M	0.37764	stem_anchor	0	False
461	I	0.37623	stem_anchor	0	False
171	S	0.37342	DI	1	False
155	T	0.37201	DI	2	False
324	V	0.37201	DIII	0	False
8	N	0.36498	DI	0	False
480	I	0.36217	stem_anchor	0	False
52	N	0.31646	DI	0	False
394	K	0.30872	DIII	0	False
225	S	0.16878	DII	0	False
457	I	0.13643	stem_anchor	0	False

E pos	majority	minority fraction	domain	#epitopes	drift vs ref
297	M	0.0865	DI	0	False
337	F	0.08368	DIII	0	False
428	V	0.06118	stem_anchor	0	False
338	S	0.05696	DIII	0	False
227	S	0.04993	DII	0	False
81	T	0.04149	DII	0	False
482	V	0.04149	stem_anchor	0	False
156	T	0.03938	DI	2	False
147	D	0.03797	DI	0	False
339	T	0.03727	DIII	0	True
382	A	0.02461	DIII	0	False
436	V	0.02461	stem_anchor	0	True
55	V	0.0225	DII	0	False
485	V	0.0211	stem_anchor	0	False
360	D	0.02039	DIII	0	False
201	M	0.01969	DII	0	False
432	V	0.01266	stem_anchor	0	False
88	A	0.01195	DII	0	True
342	E	0.01195	DIII	0	False
96	F	0.01125	DII	0	False

B.2 DENV2 (N=1577): 8 positions with minority-allele fraction $\geq 1\%$

E pos	majority	minority fraction	domain	#epitopes	drift vs ref
83	N	0.2156	DII	1	False
364	S	0.14141	DIII	0	False
232	I	0.06024	DII	0	False
310	K	0.0577	DIII	1	False
322	V	0.02536	DIII	0	True
484	V	0.02283	stem_anchor	0	True
6	I	0.01522	DI	0	True
329	D	0.01522	DIII	0	False

Section 4 watchlists (T4, results/) carry the substitution-level view.

Appendix C. Exact computational workflow

```
# data acquisition (NCBI Datasets CLI 18.37.0)
datasets summary virus genome taxon 11053 --as-json-lines > env1_ncbi_report_all.jsonl
datasets summary virus genome taxon 11060 --as-json-lines > env2_ncbi_report_all.jsonl
# filter: collection_date >= 2023-01-01, length >= 1400 nt, geography present
#   -> env{1,2}_2023_accessions.txt, env{1,2}_ncbi_report.jsonl
datasets download virus genome accession --inputfile <accessions> --include genome
# alignment, clades, mutation calls (Nextclade 3.23.0)
nextclade dataset get --name community/v-gen-lab/dengue/env1 --output-dir nc_env1
nextclade dataset get --name community/v-gen-lab/dengue/env2 --output-dir nc_env2
nextclade run --input-dataset nc_env1 --output-all ncout1 env1_2023_genomes.fna
nextclade run --input-dataset nc_env2 --output-all ncout2 env2_2023_genomes.fna
# epitope structures (RCSB PDB 4C2I, 4UIF, 4UTA, 4UT6; contacts <=6A via biopython)
python3 code/pdb_contacts.py
python3 code/fix_mapping.py && code/build_epitopes.py && code/atlas_analysis.py \
    && code/atlas_v2.py && code/stats_tests.py && code/make_figures.py && code/make_tables.py
pandoc paper/paper.md --pdf-engine=pdflatex -o paper/paper.pdf
```